Cart Abandonment Analysis

Diagnosing checkout flow drop-offs and identifying high-priority product interventions across 67.4M clickstream events.

MySQL Python (Pandas) ETL Power BI Dashboard

EXECUTIVE PROBLEM FRAMING

The Core Business Question

> **“Where in the checkout flow are customers dropping off most, and what should the product team fix first?”**

The Diagnostic Goal

Pinpoint the exact conversion friction points across categories, brands, and price tiers.

The Actionable Goal

Prioritize product, UI/UX, and payment infrastructure fixes with measurable GMV recovery.

PLATFORM OVERVIEW

Macro Conversion Scorecard

CART SESSIONS	PURCHASES	ABANDONMENT RATE	POTENTIAL LOST GMV
1.74M	648.36K	62.81%	$381.90M
1,743,342 Intent Sessions	648,364 Confirmed Orders	1.09M Abandoned Carts	Unrealized Platform Value

Over 6 out of 10 users who add items to cart abandon prior to purchase, trapping $381.9M in unrealized revenue.

FUNNEL DIAGNOSIS

Full Journey Conversion Funnel

1. Product Views
Top of Funnel



13.77M
100% Base Volume

↓ View-to-Cart Drop-off: 87.34% (Exploratory Browsing)

2. Cart Sessions
Active Intent



1.74M
12.66% of Views

↓ Cart-to-Purchase Drop-off: 62.81% (Critical Commercial Leakage)

3. Completed Purchases

End Conversion



648.36K
4.71% Full Conversion

REVENUE LEAKAGE BREAKDOWN

Category & Product Concentration

Top 5 Categories by Lost GMV

1. Electronics	**$258.37M**
2. Appliances	**$28.85M**
3. Computers	**$24.58M**
4. Auto	**$1.76M**
5. Apparel	**$1.74M**

Top 5 Product Types by Lost GMV

1. Smartphone	**$218.93M**
2. TV	**$19.55M**
3. Notebook	**$18.43M**
4. Washer	**$8.79M**
5. Clocks	**$8.33M**

Strategic Finding: Electronics generates 67.6% of platform-wide dollar losses, with smartphones alone responsible for $218.9M.

BRAND EXPOSURE

Brand-Level Revenue Exposure

Apple

#1 Leaking Brand

$160.55M

42.0% of total platform-wide lost GMV

Driven primarily by premium iPhone cart additions that hesitate during final payment checkout.

Samsung

#2 Leaking Brand

$64.24M

16.8% of total platform-wide lost GMV

Concentrated across smartphones.

Other Key Leaking Brands

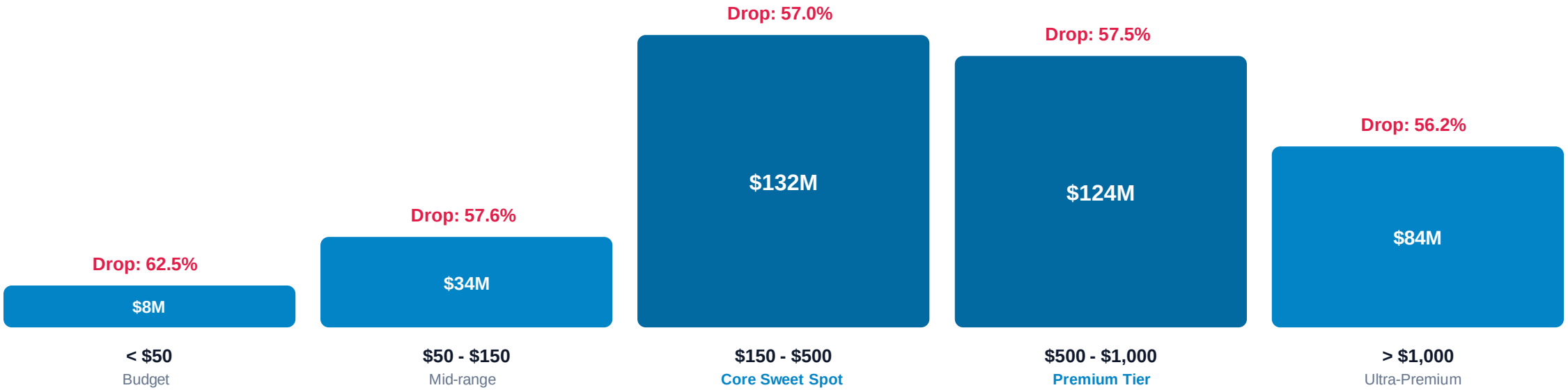
Xiaomi	**$19.26M**
LG	**$8.89M**
Huawei	**$6.21M**

Combined Apple + Samsung: 58.8% of all losses

Optimizing the checkout journey for just Apple and Samsung devices addresses nearly 60% of total revenue leakage.

PRICE SENSITIVITY ANALYSIS

Cart Abandonment & Lost GMV by Price Tier



Key Takeaway: 67% of all lost GMV is concentrated between $150 and $1,000. Price hesitation occurs at the mid-to-premium boundary .

TEMPORAL MATRIX INSIGHTS

Leakage Timing by Lost GMV: Day vs Hour

Friday & Saturday Daytime Window

Continuous band of heavy lost GMV across **Friday (Hours 02:00–13:00)** and **Saturday (Hours 04:00–08:00)**.

Shoppers populate carts during weekend planning sessions but pause prior to payment authentication.

Thursday 18:00

The heatmap displays an extreme dark navy outlier at **Thursday (Hours 18:00)**.

Methodological Note:

The heatmap matrix is calibrated on **Total Lost GMV ($)** rather than raw click frequency, preventing low-cost accessories from obscuring high-value checkout leakage windows.

ACTION CATALOG

Top Leaking SKUs (Ranked by Lost GMV)

Product ID	Brand	Product Type	Total Carts	Abandoned	Lost GMV ($)	Drop Rate
1005115	Apple	Smartphone	44,398	23,792	**$24.48M**	**53.59%**
1005105	Apple	Smartphone	15,914	7,988	**$12.43M**	**50.19%**
1005135	Apple	Smartphone	8,471	4,483	**$8.42M**	**52.92%**
1005116	Apple	Smartphone	12,244	7,583	**$8.15M**	**61.93%**
1004249	Apple	Smartphone	17,225	9,127	**$7.80M**	**52.99%**

Operational Priority: A single Apple smartphone model (SKU 1005115) represents $24.48M in lost revenue with over 23,000 abandoned carts.

DATA ENGINEERING

End-to-End Pipeline Architecture

1. Python (Pandas)

- Processed 67.5M raw records
- Removed 10 null session rows
- Purged 100,519 duplicate clicks
- Extracted 3-level taxonomy
- Saved to Parquet

2. MySQL Warehouse

- Streamed in 100k chunks
- Indexing
- CTEs
- Created aggregated tables

3. Power BI Analytics

- Executive dashboard
- Locked macro KPI cards
- Custom 320x160 matrix tooltips

Pipeline guarantees 100% mathematical integrity between database aggregates and presentation dashboard visuals.

ACTION ROADMAP

Product Interventions: What to Fix First

Priority 1: High-Ticket Financing (BNPL & No-Cost EMI)

Embed instant financing on cart screens for items >$150 to mitigate payment hesitation on Apple & Samsung tech ($224M+ loss).

Priority 2: 1-Click Express Digital Wallet Checkout

Deploy Google Pay, and biometric checkout on top leaking SKUs (SKU 1005115, 1005105) to eliminate billing form fatigue.

Priority 3: Automated Retargeting & Infrastructure Scaling

Deploy automated push/email recovery within 30–60 minutes for Friday/Saturday drops, and scale payment concurrency for Thursday spikes.

Priority 4: Upfront Pricing Transparency & Warranty Badges

Display transparent tax/shipping totals and official return seals inside the cart drawer to eliminate surprise fees at the final payment step.

EXECUTIVE SUMMARY & NEXT STEPS

Targeted Checkout Interventions Can Unlock Substantial GMV Recovery

By addressing high-ticket financing friction on Apple and Samsung smartphones, automating weekend retargeting, and streamlining checkout forms, the product team can directly recover high-value dropped carts.

$381.90M
Addressable Recovery Base

58.8%
Loss in Top 2 Brands

67.0%
Loss in $150–$1,000 Tiers

Thank You!